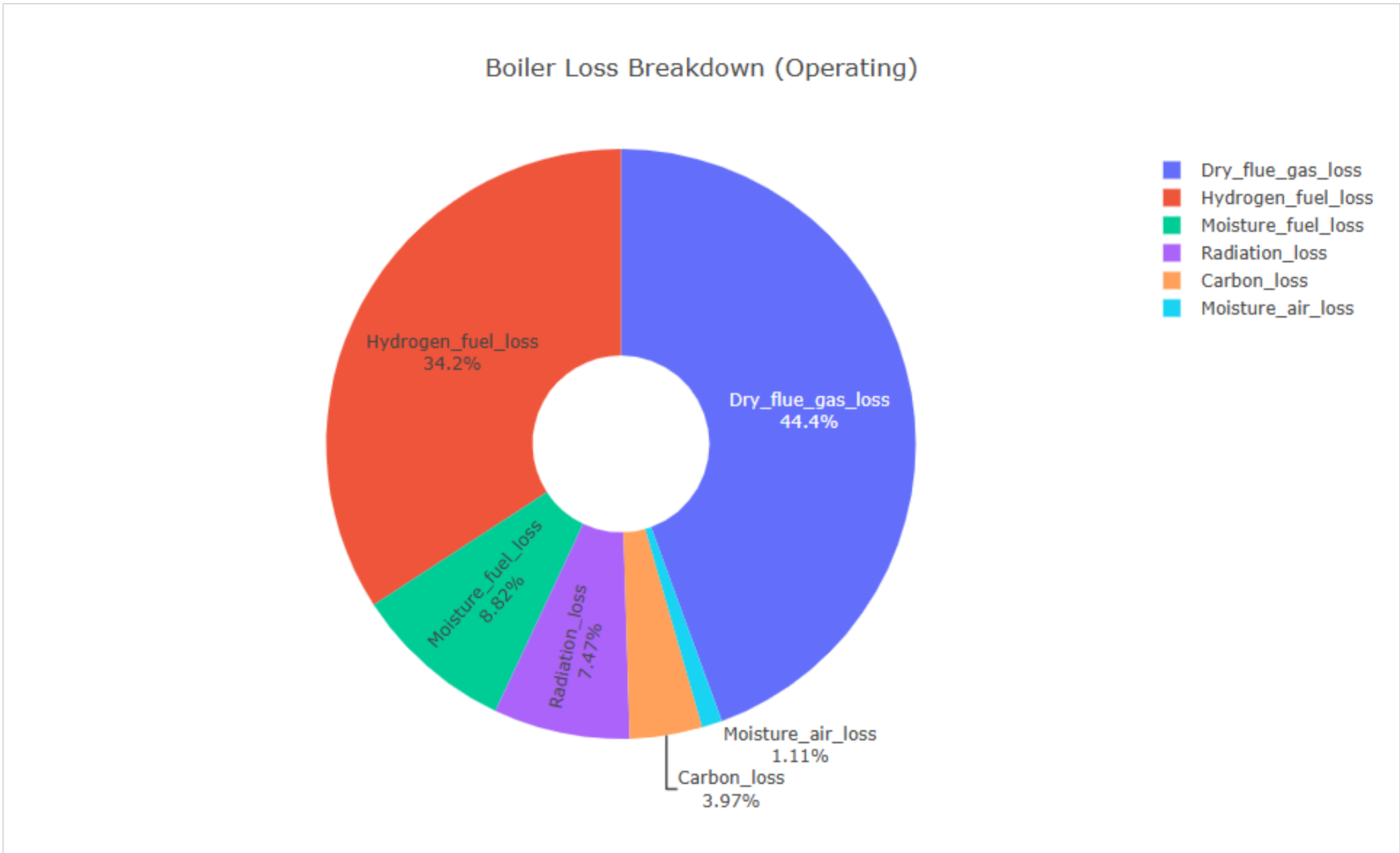
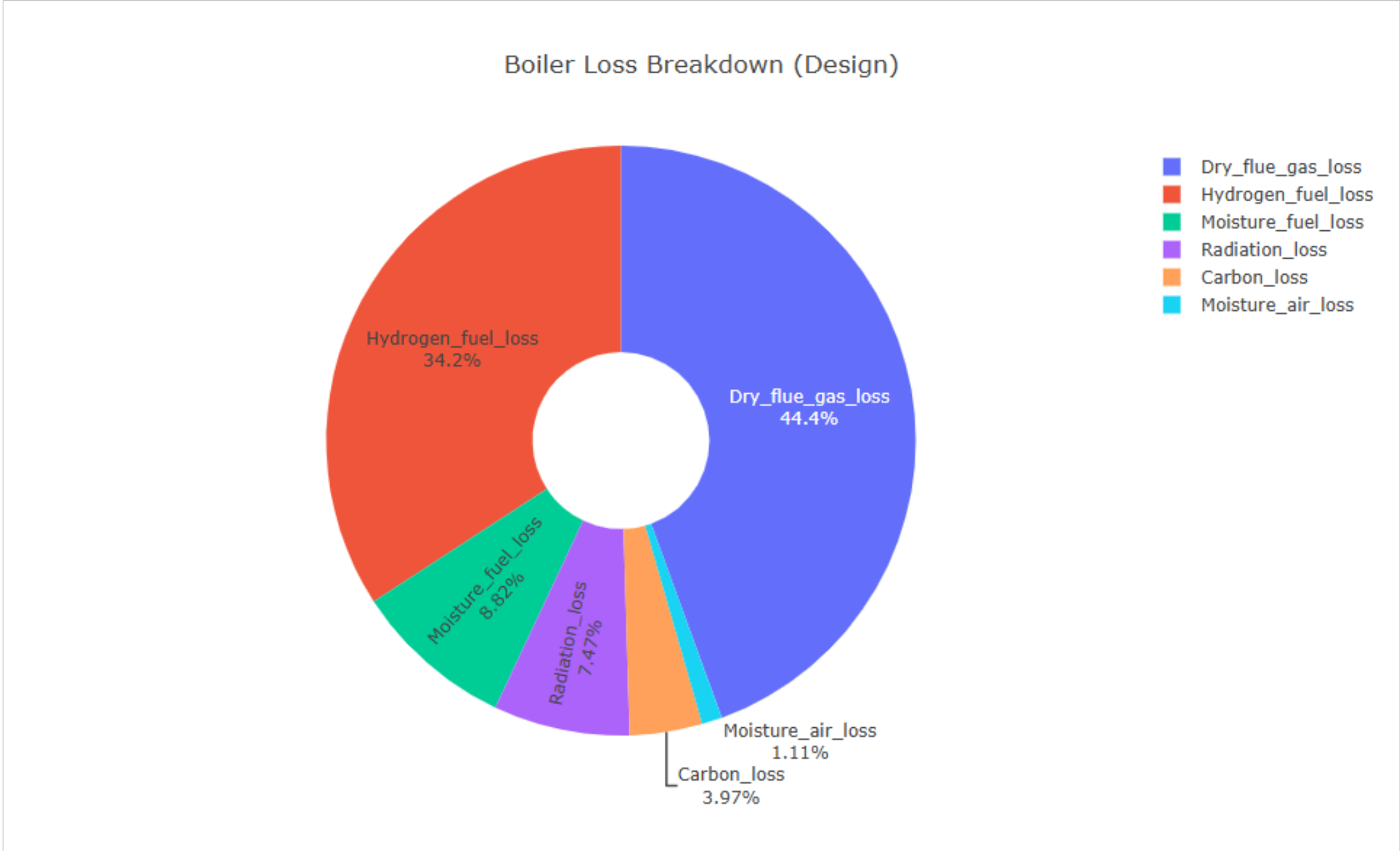


Thermal Power Plant Report

Boiler Performance

Boiler data loaded successfully.

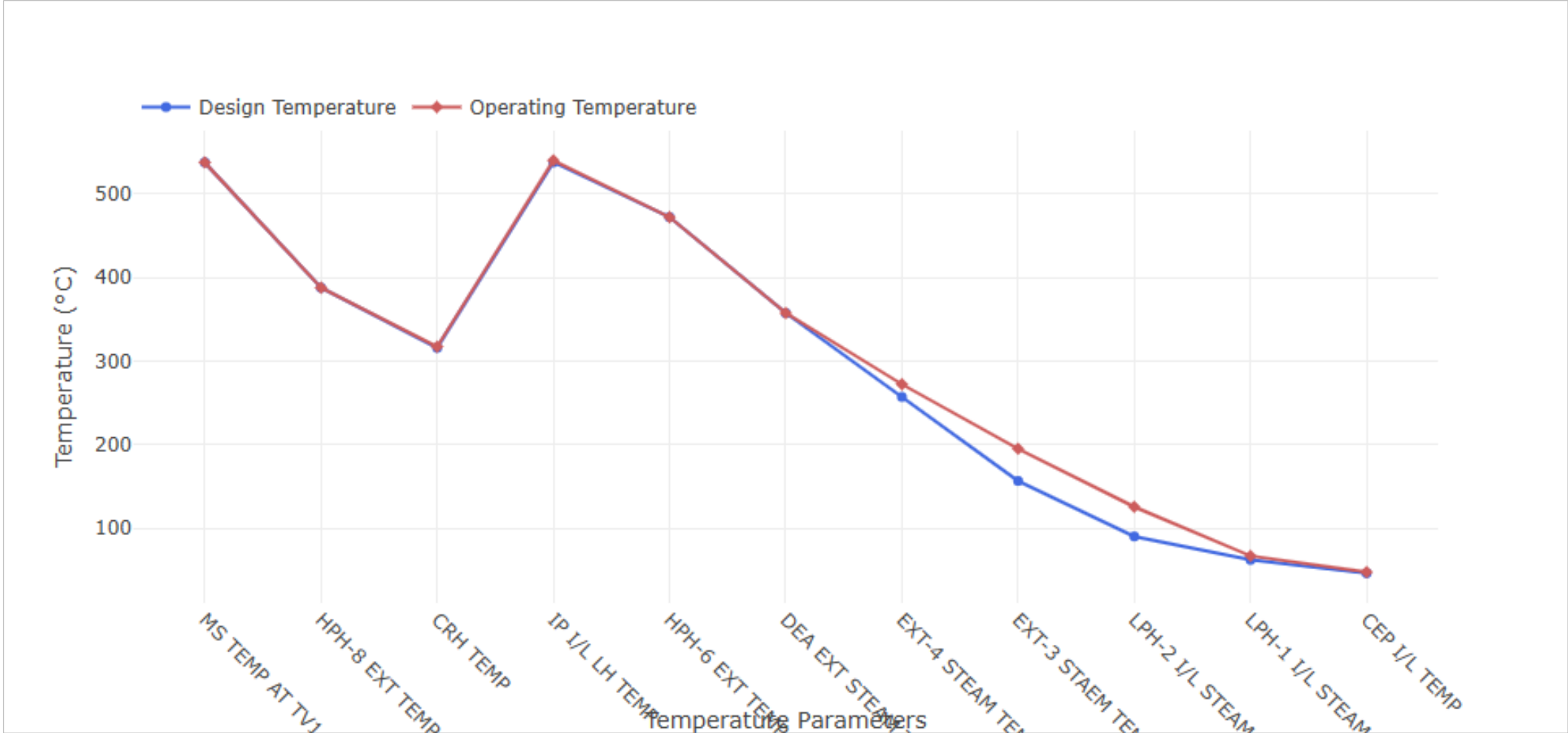


Boiler Data

Loss Type	Design (%)	Operating (%)
Dry Flue Gas Loss	5.59	5.59
Carbon Loss	0.5	0.5
Moisture Fuel	1.11	1.11
Hydrogen Fuel	4.3	4.3
Moisture Air	0.14	0.14
Radiation Loss	0.94	0.94
Total Loss	12.58	12.58
Boiler Efficiency	87.417	87.417

Turbine Temperature Profile

Temperature data is consistent across stages.

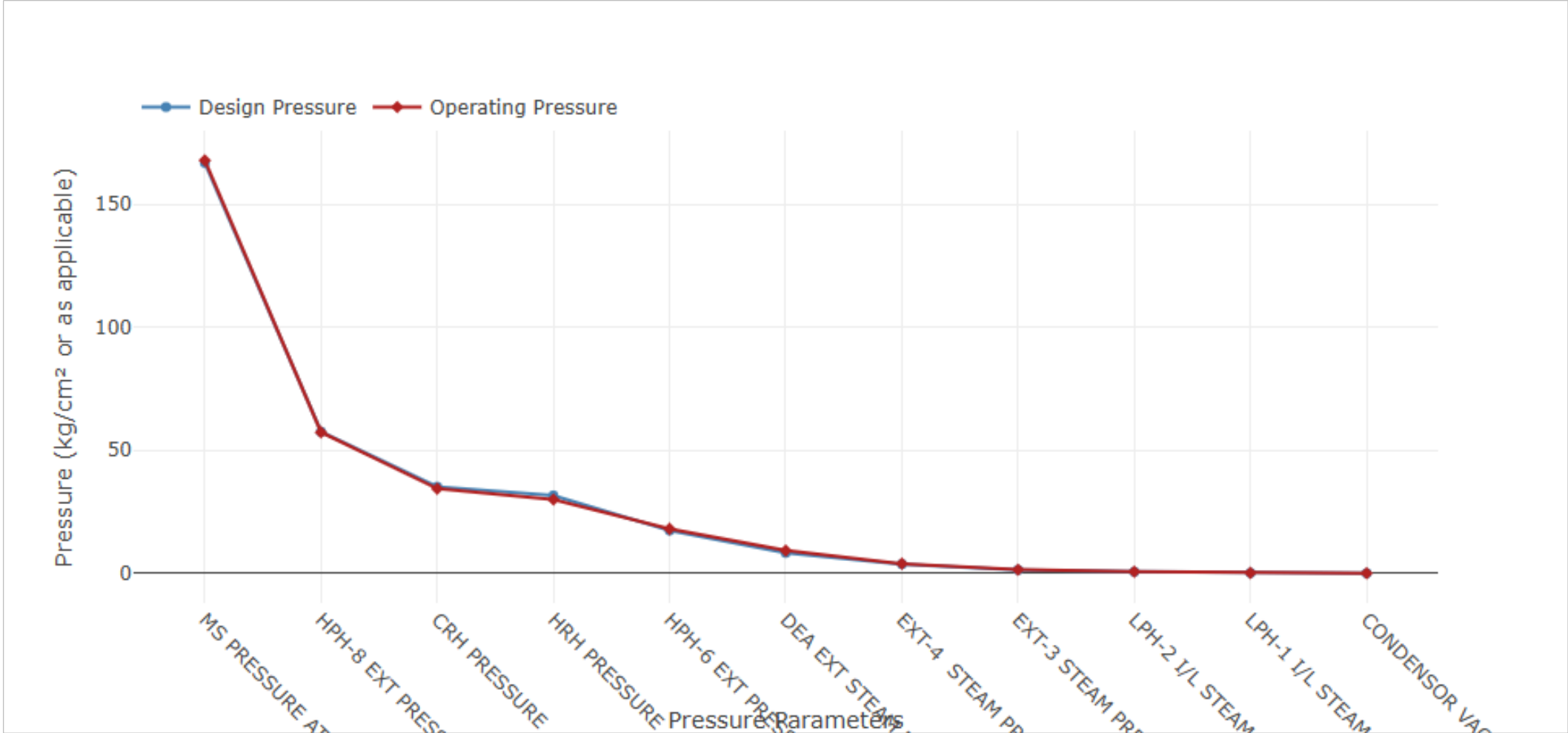


Temperature Data

Parameter	Design Value	Operating Value
MS TEMP AT TV1	538.0	538.0
HPH-8 EXT TEMP	387.92	387.92
CRH TEMP	315.8	317.17
IP I/L LH TEMP	538.0	540.36
HPH-6 EXT TEMP	472.38	472.38
DEA EXT STEAM TEMP	357.51	357.51
EXT-4 STEAM TEMP	257.09	272.17
EXT-3 STAEM TEMP	156.459	194.96
LPH-2 I/L STEAM TEMP	89.9761	125.44
LPH-1 I/L STEAM TEMP	61.87	66.61
CEP I/L TEMP	45.93	47.25

Turbine Pressure Profile

Pressure profile shows expected drop across stages.

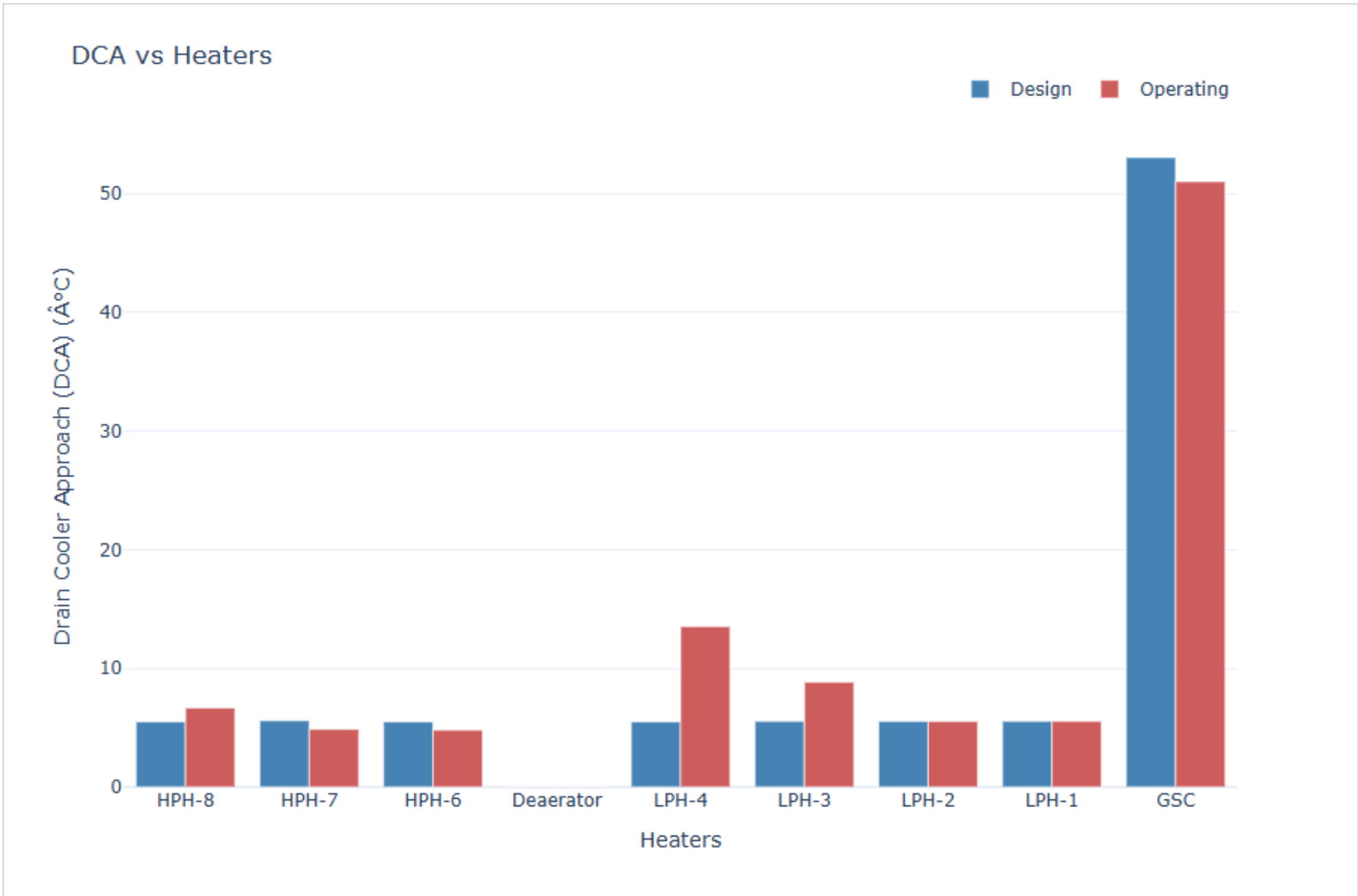
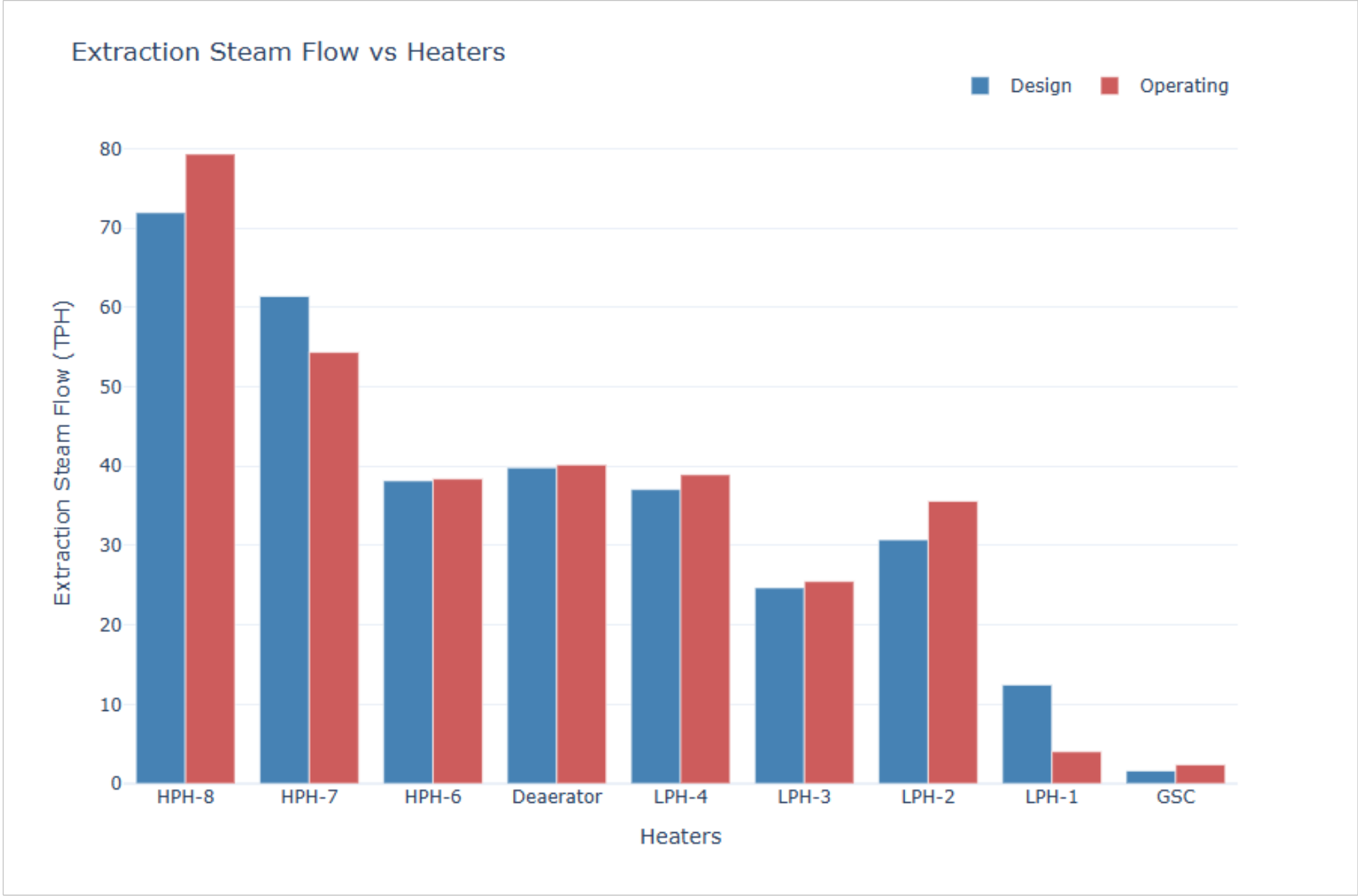


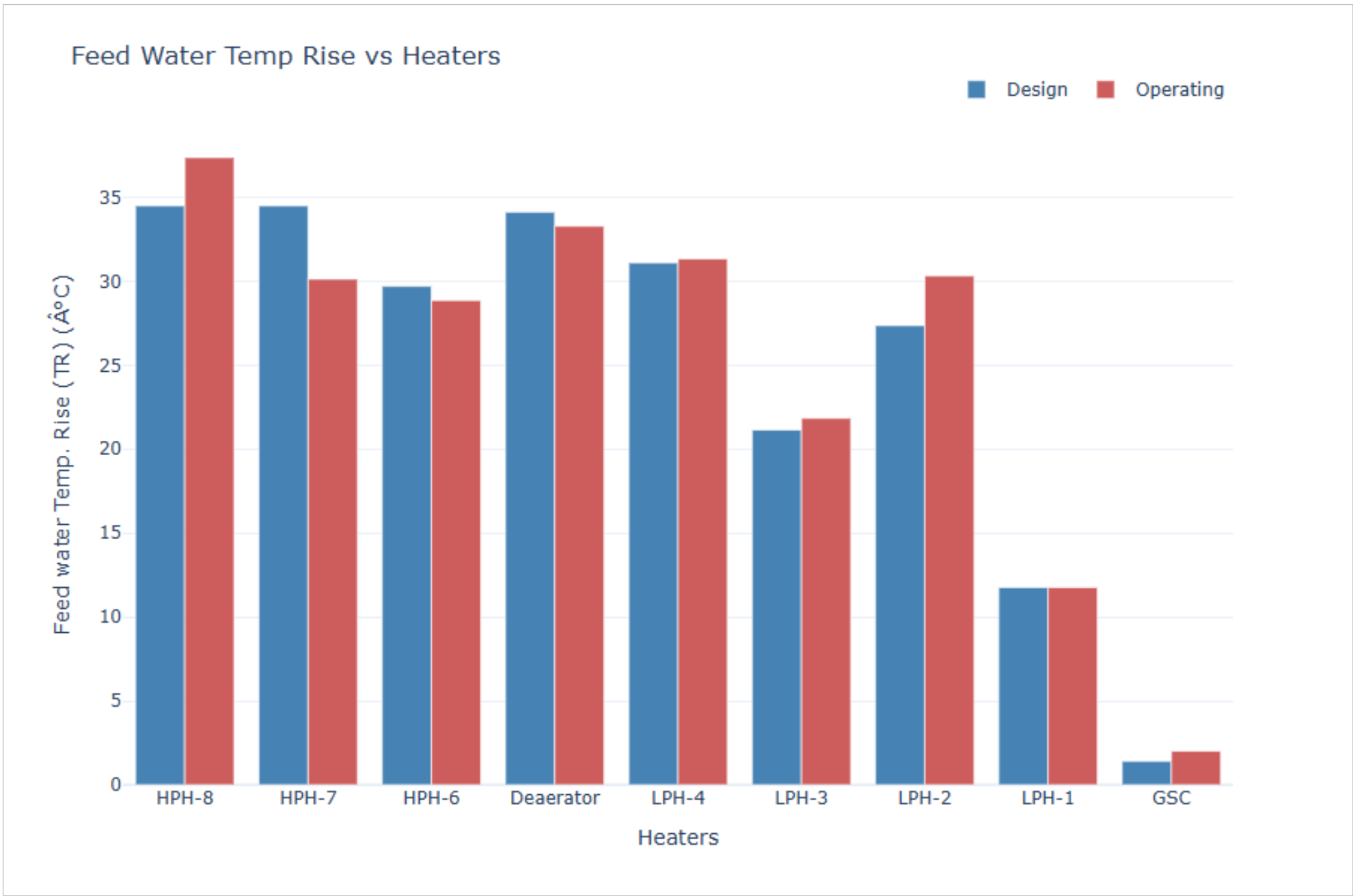
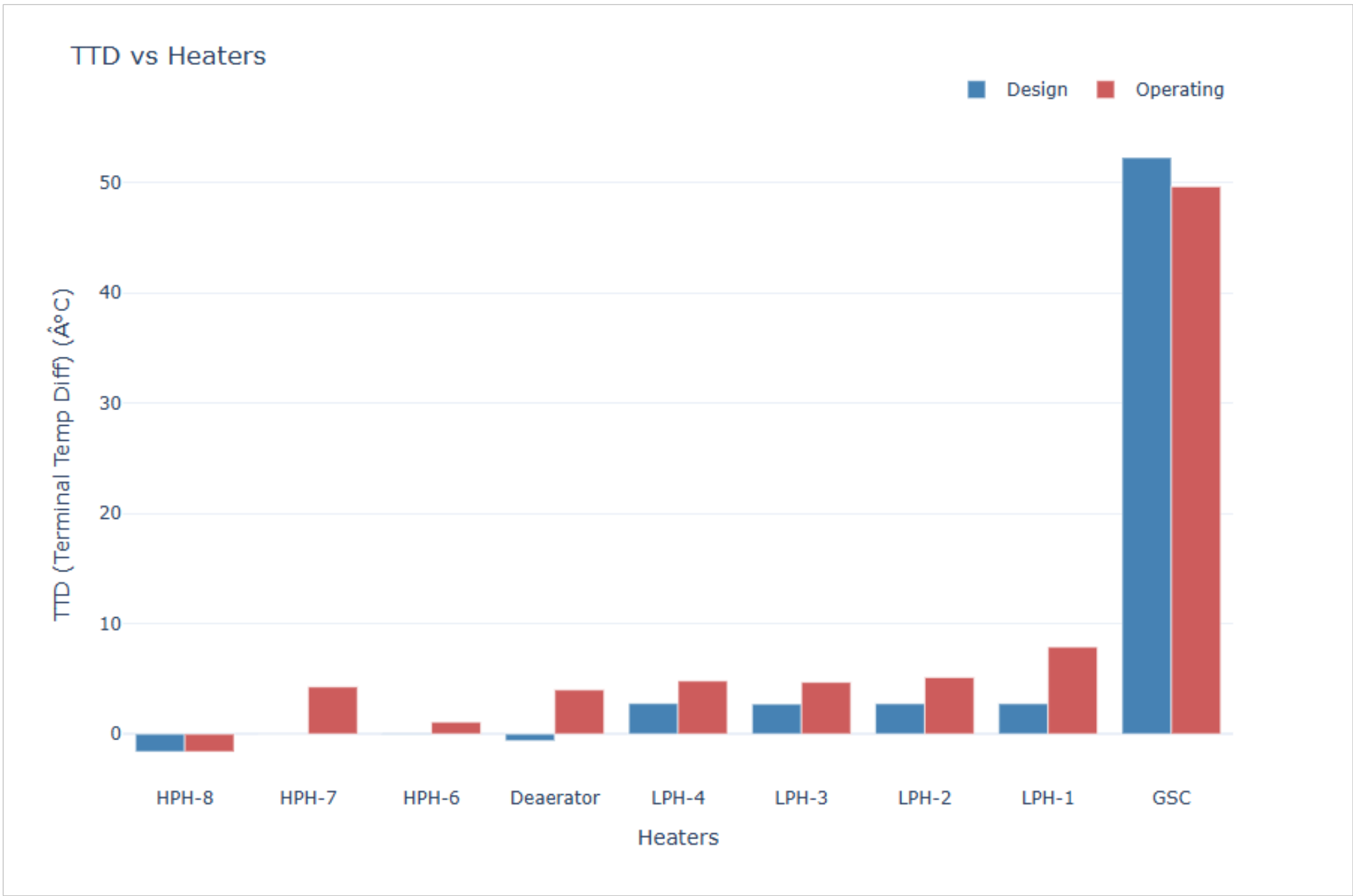
Pressure Data

Parameter	Design Value	Operating Value
MS PRESSURE AT TURBINE I/L	166.70000000000002	167.71300000000002
HPH-8 EXT PRESSURE	57.55	57.312999999999995
CRH PRESSURE	35.22	34.413
HRH PRESSURE	31.7	30.012999999999998
HPH-6 EXT PRESSURE	17.44	18.113
DEA EXT STEAM PRESSURE	8.347999999999999	9.312999999999999
EXT-4 STEAM PRESS.	3.768	3.967
EXT-3 STEAM PRESS.	1.449	1.525
LPH-2 I/L STEAM PRESS	0.6811	0.7139
LPH-1 I/L STEAM PRESS	0.2172	0.2277
CONDENSOR VACUUM (DCS)	0.10049999999999998	0.1211999999999977

Heater Performance

Heater section extracted successfully.





Heater Data																			
Description	Unit	HPH-8 D	HPH-8 O	HPH-7 D	HPH-7 O	HPH-6 D	HPH-6 O	Deaerator D	Deaerator O	LPH-4 D	LPH-4 O	LPH-3 D	LPH-3 O	LPH-2 D	LPH-2 O	LPH-1 D	LPH-1 O	GSC D	GSC O
Saturation Temp of extraction steam	°C	272.88	272.61	239.99	240.75	205.56	207.43	172.19	176.82	141.47	143.31	110.32	111.85	89.21	90.45	61.85	62.89	99.61	99.61
Feed water Temp. Rise (TR)	°C	34.5	37.37	34.5	30.13	29.7	28.86	34.12	33.28	31.1	31.34	21.14	21.84	27.36	30.32	11.76	11.76	1.41	2.01
TTD (Terminal Temp Diff)	°C	-1.62	-1.62	-0.01	4.27	0.06	1.08	-0.63	4.0	2.77	4.81	2.72	4.69	2.75	5.13	2.75	7.89	52.27	49.64
Drain Cooler Approach (DCA)	°C	5.5	6.68	5.6	4.87	5.5	4.81	0.0	0.0	5.5	13.54	5.56	8.85	5.55	5.55	5.56	5.56	53.07	51.04
Extraction Steam Flow	TPH	71.95	79.33	61.4	54.35	38.16	38.4	39.81	40.18	37.07	38.93	24.68	25.48	30.73	35.59	12.44	4.03	1.61	2.39

Heat Rate calculation

Heater section extracted successfully.

Heat Rate

Metric	Design Value	Operational Value	Unit
Feedwater flow (for intial iteration) MFWii	914138.52	939424.33	kg/hr
Calculated Feedwater flow (MFWfi)	914138.52	939424.33	kg/hr
Max Iterations	10.0	17.0	-
Heat Rate (HR)	1922.94	1988.79	kCal/kWh
Main steam flow (Ms)	914138.52	939520.74	kg/hr
Main steam(Hms)	3398.96	3397.81	kJ/kg

Condenser Performance

Condenser section extracted successfully.

Condenser

Parameter	Unit	Design	Operating
Condenser DP of tube side	kPa(a)	84.5	84.5
Condenser Effectiveness	%	82.7	62.56
TTD	°C	2.0	5.71
Condenser heat load steam side	MJ/h	1340.78	1404.78
Cooling water flow	TPH	33574.44	35177.16
LMTD	°C	5.44	9.71
Subcooling temperature	°C	0.0	2.37